

An accurate recognition and localization algorithm for strawberry picking robots

Jiqing Chen

College of Mechatronic
Engineering, Guangxi University,
Nanning 530007, China

Lixiang Huang

College of Mechatronic
Engineering, Guangxi University,
Nanning 530007, China

Yang Huang

College of Mechatronic
Engineering, Guangxi University,
Nanning 530007, China

ABSTRACT

With the development of smart agriculture, automatic picking robots have become a research hotspot and achieving accurate recognition of fruits and low-cost deployment are the key technologies. A lightweight strawberry detection model YOLOv10-SD (strawberry detector) is proposed to address the problem that strawberry targets are small and susceptible to occlusion by branches and leaves. Firstly, StarNet is used to reconstruct the backbone network of YOLOv10n to reduce the number of parameters and achieve fast detection. Then, the Coordinate Attention (CA) module is introduced to enhance the accurate recognition of the region of interest or the position of important objects in the convolutional neural network, thereby improving the detection effect of strawberries under complex backgrounds. Finally, this model integrates Inner-IOU into CIOU as the loss function, accelerates the regression process of low IOU samples through a larger auxiliary bounding box, and is conducive to enhancing the detection ability of small target strawberries. Experiments are carried out on a strawberry dataset under a natural growth environment. The results show that compared with the current mainstream object detection algorithms YOLOv5n, YOLOv7-tiny, YOLOv8n, YOLOv9-c and YOLOv10n, YOLOv10-SD exhibits better comprehensive performance in terms of model parameter quantity, real-time performance and detection accuracy. The YOLOv10-SD model proposed in this study improves the ability to identify and locate strawberries in natural environments while ensuring lightweight, providing a technical reference for efficient smart strawberry picking.

Keywords: Strawberry, Small target, Automatic picking robot.

INTRODUCTION

Strawberries are rich in various vitamins and minerals needed by the human body and have extremely high nutritional value. They are one of the fruits that are deeply loved by consumers [1]. In recent years, with the continuous expansion of strawberry cultivation, China has become the country with the largest strawberry planting area and output in the world [2]. However, the current strawberry picking still mainly relies on manual operation. Due to the short ripening period of strawberries and the fruit's fragility, the manual picking process is labor-intensive and inefficient, and the picking cost increases year by year, which seriously restricts the further development of the industry [3]. Therefore, improving the automation level of strawberry picking and reducing labor costs have become important issues that need to be solved in this field.

With the development of artificial intelligence, object detection algorithms based on deep learning have gradually received widespread attention. For example, Xiaoqiang Du[4] modified the YOLOv7 model, accelerated the convergence speed of the network, and achieved accurate detection of ripe strawberries and their degree of occlusion. Yanchao Zhang[5] The number of layers of YOLOv4-tiny was reduced, and the optimized network achieved real-time detection of strawberries, with a speed increase of 25.93% while the accuracy decreased by 0.62%. Yang Yu[6] proposed a rotation of YOLO with a lightweight network instead of a convolutional neural network as the backbone network, which showed good real-time performance. Although the above methods have achieved good results, the increase in detection speed is often at the expense of reduced detection accuracy. It is still difficult to achieve high-precision real-time detection in complex environments.

Aiming at the above problems, this paper proposes a lightweight deep learning model YOLOv10-SD, which improves the speed without affecting the detection accuracy, realizes accurate and real-time identification and localization of strawberries, and provides valuable references for the realization of efficient and accurate automated picking operations.

MATERIALS AND METHODS

1.1 Dataset Construction and Processing

The strawberry images were collected from a professional strawberry plantation in Nanning, Guangxi Zhuang Autonomous Region. A total of 2000 images were collected in this shooting, and each image was adjusted to 640×640 pixels to construct the original image dataset for strawberry maturity detection. The dataset was randomly divided into training, validation, and test subsets in a ratio of 8:1:1.

This study divides the different maturity of strawberries into three levels based on the peel color and market demand: high ripeness, medium ripeness and low ripeness. To improve the robustness and generalization ability of the model, five image processing techniques were used for data enhancement, namely rotation, brightness adjustment, Gaussian noise addition, contrast adjustment and image occlusion. These methods were flexibly combined and applied to enhance the data set and finally generated 4,000 training sets, 500 validation sets and 500 test sets.

1.2 YOLOv10 network structure

YOLOv10[7] was proposed by the Tsinghua University team. It inherits the real-time and accurate features of the YOLO series and optimizes and improves the network structure, training process, and feature extraction capabilities. YOLOv10 is mainly composed of input, backbone, neck, and head.

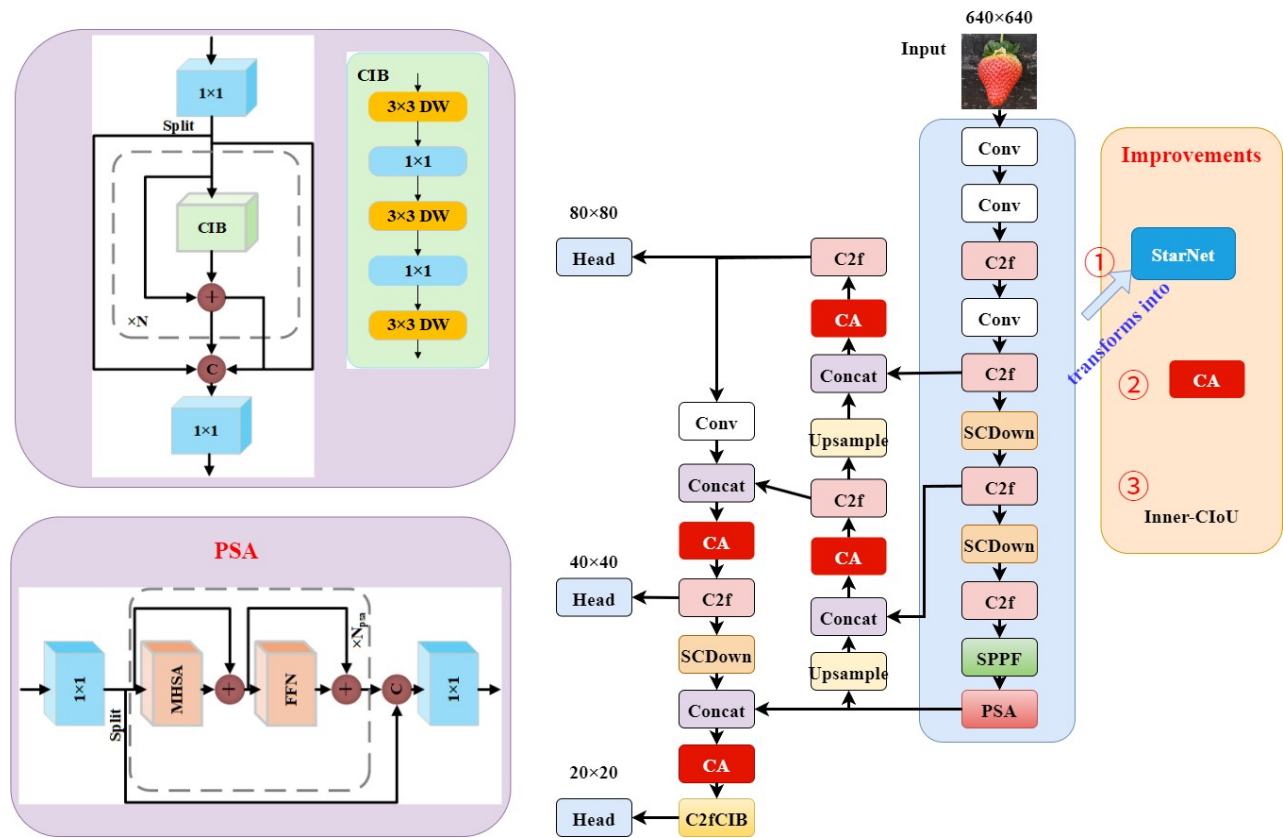


Figure 1. The overall framework diagram of YOLOv10-SD includes three improvements: ① Replace the backbone network with StarNet; ② Embed the CA attention mechanism in Neck; ③ Use Inner-CIoU as the loss function.

1.3 YOLOv10 network improvement

This study improved YOLOv10n and proposed a strawberry detection model of different maturity levels, called YOLOv10n-SD. The model architecture is illustrated in Fig.1.

Although the overall efficiency-accuracy driven model design strategy of YOLOv10 achieves efficient parameter utilization, there is still room for improvement. Since StarNet[8] demonstrates excellent performance and low latency

with a compact network structure and efficient budget, this study utilizes StarNet to construct the backbone network. StarNet follows the traditional hierarchical network with a 4-stage architecture, as shown in Fig.2. In which the resolution is directly downsampled with convolutional layers and the number of channels in each stage is doubled. Features are extracted by repeating through multiple Star Blocks. The different sizes of StarNet are determined by the number of input embedding channels and the number of blocks of Star Blocks. In this study, $\{n_1 = 1, n_2 = 1, n_3 = 3, n_4 = 1\}$ was set to ensure a balance between lightweight and accuracy.

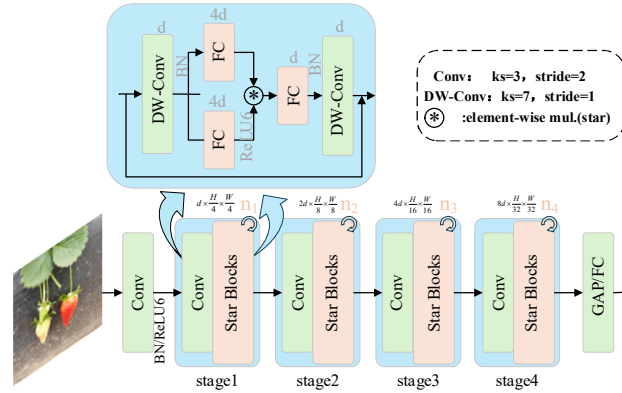


Figure 2. StarNet architecture diagram

To improve the model's ability to recognize strawberries under complex conditions, this study adopts the Coordinate Attention (CA) mechanism[9] to enhance the sensitivity of the neural network to spatial position information. The implementation of the CA mechanism is shown in Fig.3. Firstly, two 1D vectors are obtained by average pooling of X-axis and Y-axis of the feature map. After concatenating the spatial dimensions, the channels are compressed by convolution and BN and Non-linear operations are employed to encode the spatial information in the vertical and horizontal directions. Subsequently the features are further separated and processed through two convolutional layers to obtain the attention weights for each spatial dimension independently. Finally, normalization and weighting make the model more accurate in locating to and identifying the target area.

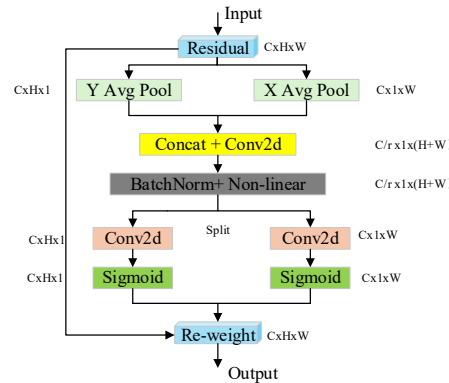


Figure 3. Schematic comparison of the Coordinate Attention

Given the current problem of insufficient generalization of IoU-based bounding box regression loss, this study utilizes the idea of Inner-IoU [10] to transform CIoU into Inner-CIoU as the loss function of the model, as presented in Equation 1. Essentially, Inner-CIoU introduces an auxiliary bounding box based on CIoU to compute the IoU loss, as illustrated schematically in Fig. 4. The scaling size of the auxiliary bounding box used to calculate the loss is controlled by the scaling factor ratio. For the strawberry dataset, there are many small targets and occluded targets. The scaling factor is greater than 1, that is, the original Anchor Box and Target Box are enlarged, which is beneficial to improve the detection effect and effectively accelerate the bounding box regression process. In this experiment, the scaling factor was set to 1.15.

$$L_{Inner-CIoU} = L_{CIoU} + IoU - IoU^{inner} \quad (1)$$

Where IoU is the intersection of Anchor Box and Target Box, and IoU^{inner} is the intersection of InnerAnchor Box and InnerTarget Box.

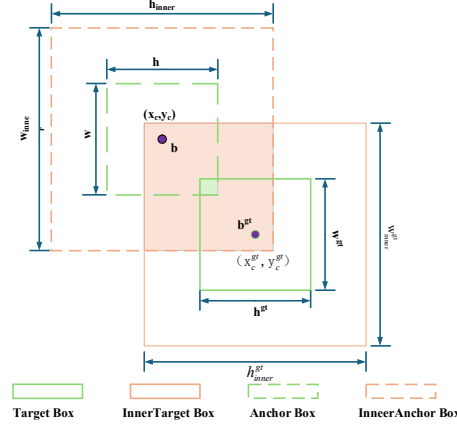


Figure 4. Description of Inner-IoU

EXPERIMENTS AND RESULTS

1.4 Evaluation Metrics

This study adopts multiple evaluation indicators to comprehensively evaluate the performance of the proposed model, including mean Average Precision (mAP), model parameters (Params), and detection frame rate (FPS). These indicators together form a comprehensive evaluation framework to quantify the effectiveness of the model in practical applications and its resource consumption.

$$P = \frac{TP}{TP + FP} \quad (2)$$

$$R = \frac{TP}{TP + FN} \quad (3)$$

$$AP = \int_0^1 PRdR \quad (4)$$

$$mAP = \frac{1}{n} \sum_{i=1}^n AP \quad (5)$$

Where: TP represents the number of samples predicted by the model as positive and correct; FP represents the number of samples predicted by the model as positive but turned out to be negative; FN represents the number of samples predicted by the model as negative but turned out to be positive; n represents the number of data categories, here n=3.

1.5 Ablation Experiment

Ablation experiments were carried out to verify the effectiveness of each improvement point, and the results are listed in Table 1. After the lightweight transformation of the backbone using StarNet, the parameter volume of the model is reduced by 20.7% to 1.80 M. Thanks to the efficient feature representation achieved by StarNet without increasing the computational complexity and complicated network design, the mAP is slightly reduced by 0.5%. To achieve model performance improvement with low computational overhead the CA attention mechanism is introduced, which can capture both cross-channel information and direction and position sensing information to achieve accurate localization and identification of the detected target. After embedding the CA attention mechanism in the Neck network, the accuracy drop caused by lightweight is significantly alleviated, with P, R and mAP increasing by 2.5, 1.7 and 1.6 per cent

respectively. An increase in recall rate means a reduction in missed detections. The addition of the attention mechanism only increases the number of model parameters by 0.03M. By replacing the model's loss function with Inner-CIoU, the model detection effect is further improved with the assistance of auxiliary bounding boxes without increasing the model size. The final improved model YOLOv10n-SD increases P, R and mAP by 2.6%, 2.9% and 2.1%, respectively, and decreases Params by 19.38% compared to YOLOv10n.

Table 1 Results of improved YOLOv10n ablation experiment

StarNet	CA	Inner-CIoU	P/%	R/%	mAP/%	Params/M
×	×	×	89.4%	86.3%	92.2%	2.27
√	×	×	88.3%	85.9%	91.7%	1.80
√	√	×	90.8%	87.6%	93.3%	1.83
√	√	√	92.0%	89.2%	94.3%	1.83

1.6 Comparison of different detection algorithms

This study conducts comparative experiments on the proposed algorithm and several mainstream object detection algorithms to verify the superiority of YOLOv10n-SD. The experimental results are shown in Table 2. YOLOv10n-SD has the least storage usage, and the number of parameters is only 1.83 M. Compared with the lightweight networks YOLOv5n and YOLOv8n, the mAP of YOLOv10n-SD increased by 4.7% and 2.8 respectively. The number of parameters of YOLOv10n-SD is only 30.4% and 7.2% of that of YOLOv7-tiny and YOLOv9-c respectively, but the mAP increased by 2.6 and 1.2 per cent respectively. In terms of detection speed, YOLOv10n-SD achieved a detection speed of 281 f/s, which is 28.3%, 51.9%, 36.4%, 170.2% and 26.0% higher than YOLOv5n, YOLOv7-tiny, YOLOv8n, YOLOv9-c and YOLOv10n respectively. Comprehensive comparison results show that YOLOv10n-SD not only has advantages in detection accuracy, but also demonstrates excellent real-time performance and potential for deployment on mobile devices, which is suitable for scenarios with limited computing resources.

Table 2 Results of the comparative experiments.

No	Models	P	R	mAP%	Params	FPS
1	YOLOv5n	87.9%	85.1%	89.6%	2.50	219
2	YOLOv7-tiny	89.5%	86.4%	91.7%	6.02	185
3	YOLOv8n	89.1%	85.9%	91.5%	3.01	206
4	YOLOv9-c	90.3%	88.0%	93.1%	25.32	104
5	YOLOv10n	89.4%	86.3%	92.2%	2.27	223
6	YOLOv10n-SD	92.0%	89.2%	94.3%	1.83	281

From the comparison of the detection results in Fig. 5, it is intuitively observed that YOLOv9-c, YOLOv10n, YOLOv7-tiny, YOLOv8n, and YOLOv5n exhibit leakage detection, and especially perform poorly in recognizing small-sized and partially occluded strawberries. In contrast, the improved model YOLOv10n-SD proposed in this study displayed significant advantages in several aspects.

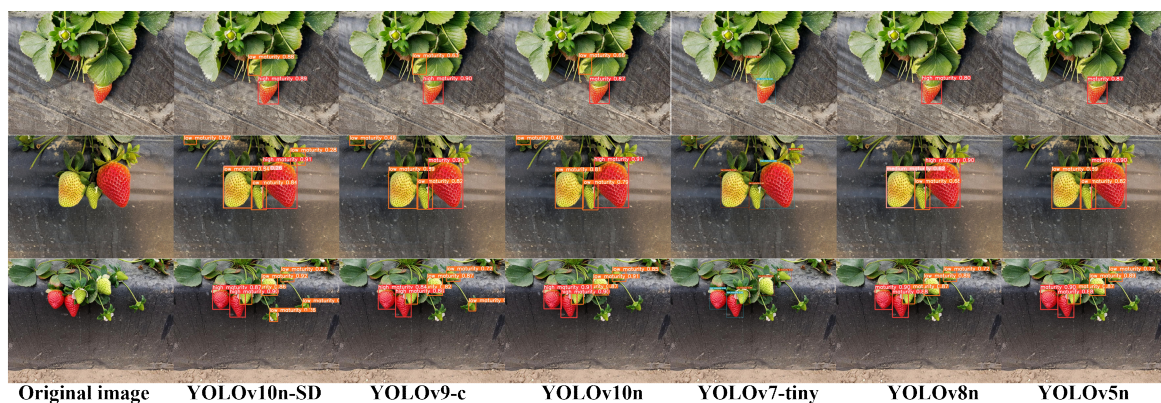


Figure 5. Comparison of the effects of YOLOv10-SD and mainstream models

CONCLUSION

In this paper, a high-precision and lightweight strawberry maturity object detection model YOLOv10n-SD is proposed. YOLOv10n is improved through StarNet, CA mechanism and Inner-CIoU loss function, which effectively solves the problem of low detection accuracy caused by complex factors such as branch and leaf occlusion and small targets. Experimental results show that YOLOv10-SD has a P of 92.0%, an R of 89.2%, a mAP of 94.3%, a Params of 1.83 M, and an FPS of 281 f/s. Compared with the base model YOLOv10n, YOLOv10n-SD achieves a significant improvement in the detection effect, real-time performance, and resource consumption. YOLOv10n-SD also offers significant competitiveness compared with the current mainstream object detection algorithms.

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